Team Solutions

1. The correct answer will have digits in increasing order, so a strategy would be to look for the largest single digit factor of 96 (8) and iterate (left with 12, largest factor is 6, leaving 2). The answer is $\boxed{268}$.

2. Let $n = 2^a 3^b 5^c$; then $m = 10n = 2^{a+1}3^b5^{c+1}$. Note that n has $(a+1)(b+1)(c+1)$ factors, and m has $(a+2)(b+1)(c+2)$ factors. The difference is $(b+1)(a+c+3) = 21$. Arguably, c should be zero to minimize the value of n, so $(b+1)(a+3) = 21$ has non-negative integer solutions of $(a,b) = (18,0)$, $(4,2)$, and $(0,6)$. The smallest of these is the middle, which is equal to $2^4 3^2 = \boxed{144}$.

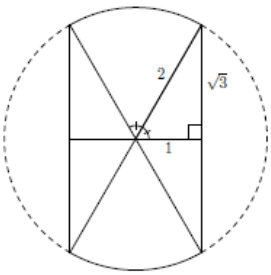
3. Let the legs of the triangle be x and y, then $xy = 12$, so:

$$\sin 2A = 2\sin A \cos A = 2\left(\frac{x}{10}\right)\left(\frac{y}{10}\right) = \frac{xy}{50} = \boxed{\frac{6}{25}}.$$

4. The sum of the elements in T is 4905. The first palindrome less than 4905 is 4884, so $\boxed{21}$ has been removed from the set.

5. Note that the sum of the pairwise sums is 4 times the sum of the elements in the sequence, so $A+B+C+D+E = 59$. Note also that $A+B = 18$ and $D+E = 29$, so $C = 12$. The second smallest sum is $A+C$, so $A = 8$ and therefore $B = 10$. By similar reasoning, $C+E = 28$, so $E = 16$ and $D = 13$, so the answer is $\boxed{(8,10,12,13,16)}$.

6. The cross section of this cylindrical region looks like a circle of radius 2 with sectors cut off by the chords $x = 1$ and $x = -1$. Thus we can compute its area by breaking it into 30-60-90 triangles and 60° circular wedges. Thus the surface area of the of top and bottom of the solid is $2\sqrt{3} + \frac{4\pi}{3}$. The sides consist of two rectangles of height 8 and width $2\sqrt{3}$, and two curved walls of height 8 and width $\frac{2\pi}{3}$. Thus the surface area is $2\left(2\sqrt{3} + \frac{4\pi}{3}\right) + 2 \times 16\sqrt{3} + 2 \times \frac{16\pi}{3} = \boxed{36\sqrt{3} + \frac{40\pi}{3}}$.



7. The only way that none of the four integers can differ by a multiple of 5 is if they all have different residues modulo 5, and these residues must sum to a multiple of 5. The only way this is possible is if the residues are 1, 2, 3, and 4. There are $4! = 24$ ways to arrange the

residues, then we are left with 7 "blocks" of five to be distributed among the four numbers, of which there are $C(10,7) = 120$ ways to distribute, so the answer is $\boxed{2880}$.

8. Let A, B, and C be the largest, middle and smallest number rolled. By symmetry, we know that $E[A] = 9 - E[C]$ and that $E[B] = 4.5$, and the expected value of $\underline{A}\,\underline{B}\,\underline{C}$ is

$$100E[A] + 10E[B] + E[C],$$

so the challenge is to determine the expected value of the largest die roll. This is $\sum_{k=0}^{9} kP(A = k)$, where

$$P(A = k) = P(x_1 \le k)\,P(x_2 \le k)\,P(x_3 \le k) - P(x_1 \le k-1)\,P(x_2 \le k-1)\,P(x_3 \le k-1),$$

which is

$$\frac{(k+1)^3 - k^3}{1000} = \frac{3k^2 + 3k + 1}{1000}.$$

The expected value of A is 6.975, so the expected value of the largest possible three digit number formed is $697.5 + 45 + 2.025 = 744.525$. We are looking for the integer portion, which is $\boxed{744}$.

9. $\dfrac{\log(2a-7)}{\log a} \cdot \dfrac{\log a}{\log(a-2)} = -\dfrac{1}{2} \to \log(2a-7) = -\dfrac{1}{2}\log(a-2) \to 2a - 7 = \dfrac{1}{\sqrt{a-2}}$

Let $x = \sqrt{a-2} \to 2x^2 - 3 = \dfrac{1}{x} \to 2x^3 - 3x - 1 = 0 \to (x+1)(2x^2 - 2x - 1) = 0 \to x = \dfrac{2 \pm \sqrt{12}}{4} \to x = \dfrac{1+\sqrt{3}}{2} \to \sqrt{a-2} = \dfrac{1+\sqrt{3}}{2} \to a = 2 + \left(\dfrac{1+\sqrt{3}}{2}\right)^2 \to a = 2 + \dfrac{4+2\sqrt{3}}{4} \to a = \boxed{3 + \dfrac{\sqrt{3}}{2} = \dfrac{6+\sqrt{3}}{2}}$.

10. Since the left hand side of the equation is between 0 and 1, we know that $7 \le x \le 8$, so let $x = 7 + y$, where $y = \{x\}$ is between 0 and 1. So $y^2 = (7+y)^2 - 56 \to 0 = 49 + 14y - 56 \to y = 0.5 \to x = \boxed{7.5}$.